



CSC Computing environment

Machines, environments and workloads



Lecture agenda

What is High-performance computing?

What is a supercomputer?

What are supercomputers for?

CSC's supercomputers and services

SSH Keys

Logging into Mahti

What is High-performance computing?

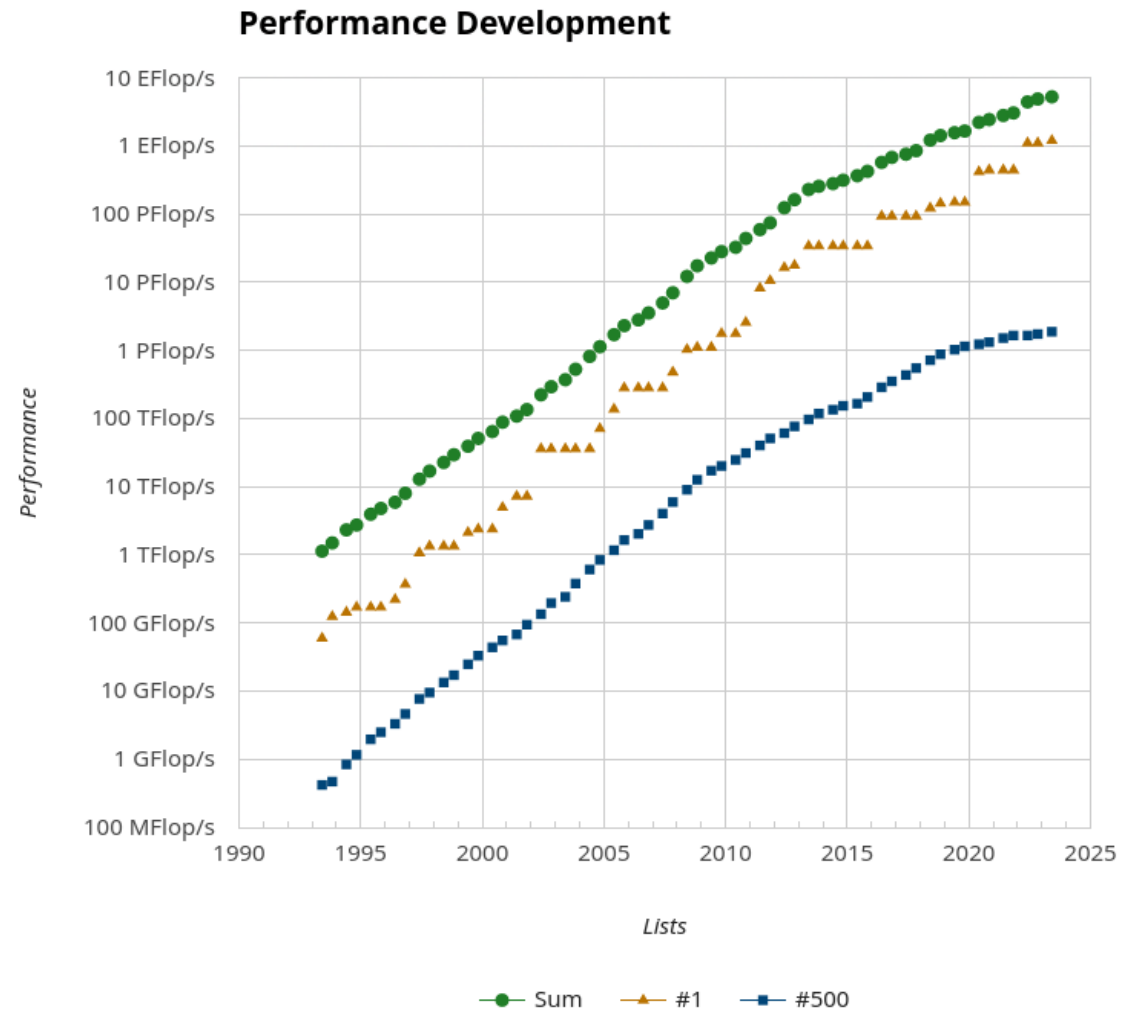
High-Performance computing

- A Field of science that includes software performance, scalability, parallelism, etc.
- Typical fields where HPC is highly relevant:
 - Weather forecasting
 - Molecular dynamics
 - Glacier research
 - Etc...
- Anything that involves long timescales, chaotic systems, simulating thousands of particles at a time etc.

High-Performance computing

- Utilizing computing power that is much larger than available in a typical desktop computer
- Performance of HPC systems (i.e. supercomputers) is often measured in floating point operations per second (flop/s)
 - For software, other measures can be more meaningful
- Currently, the most powerful system reaches $> 10^{18}$ flop/s (1 Eflop / s)

The Top 500 list

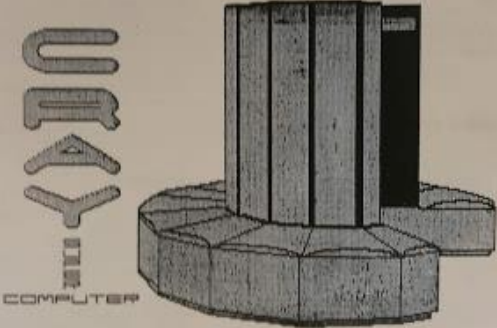


What is a supercomputer?

What does a supercomputer look like?

**CRAY X-MP -
The most powerful computer in Scandinavia**

The - CRAY X-MP/28 - supercomputer was installed in Trondheim as a result of an initiative taken by The Norwegian Institute of Technology (NTH) and SINTEF. The computer was financed by NTNE, NAVF, NTH and SINTEF and is supported by economical guaranties from Statoil and Norsk Hydro. The Computing Center at the University of Trondheim (RUNIT) is responsible for the operation of the CRAY.



COMPUTER

The CRAY X-MP/28 in Trondheim is the most powerful computer in Scandinavia. The computer was installed 1. January 1987 and is accessible 24 hours a day with operator services during daytime (08-21).

CPU's:	2	System software:	COS operating system (Later UNICOS)
Main memory:	8 megawords & 64 bits		FORTRAN, PASCAL, (C)
Cycle time:	2.3 - nanoseconds	Workstations:	Development tools for vector/parallel programs
Buffer memory:	4 m. gawords & 64 bits	Communication:	Apollo, DEC, (SUN, IRIS)
Mass storage:	10 gigabytes	Performance:	Hyperchannel A and B
Frontends:	VAX 8600, R200	- (sustained) :	50-100 Megaflops
	Sperry 1100/72	- (peak):	470 Megaflops
	IBM 4361		
	(ND 500)		

(Megaflops = millions of floating point operations per second.)



What does a supercomputer look like?



Supercomputer design

- Massively parallel design
- Hundreds or thousands of individual compute nodes
- Each node can be thought of as an individual computer and each one has its own:

- CPUs (with tens to hundreds of cores each)



- RAM



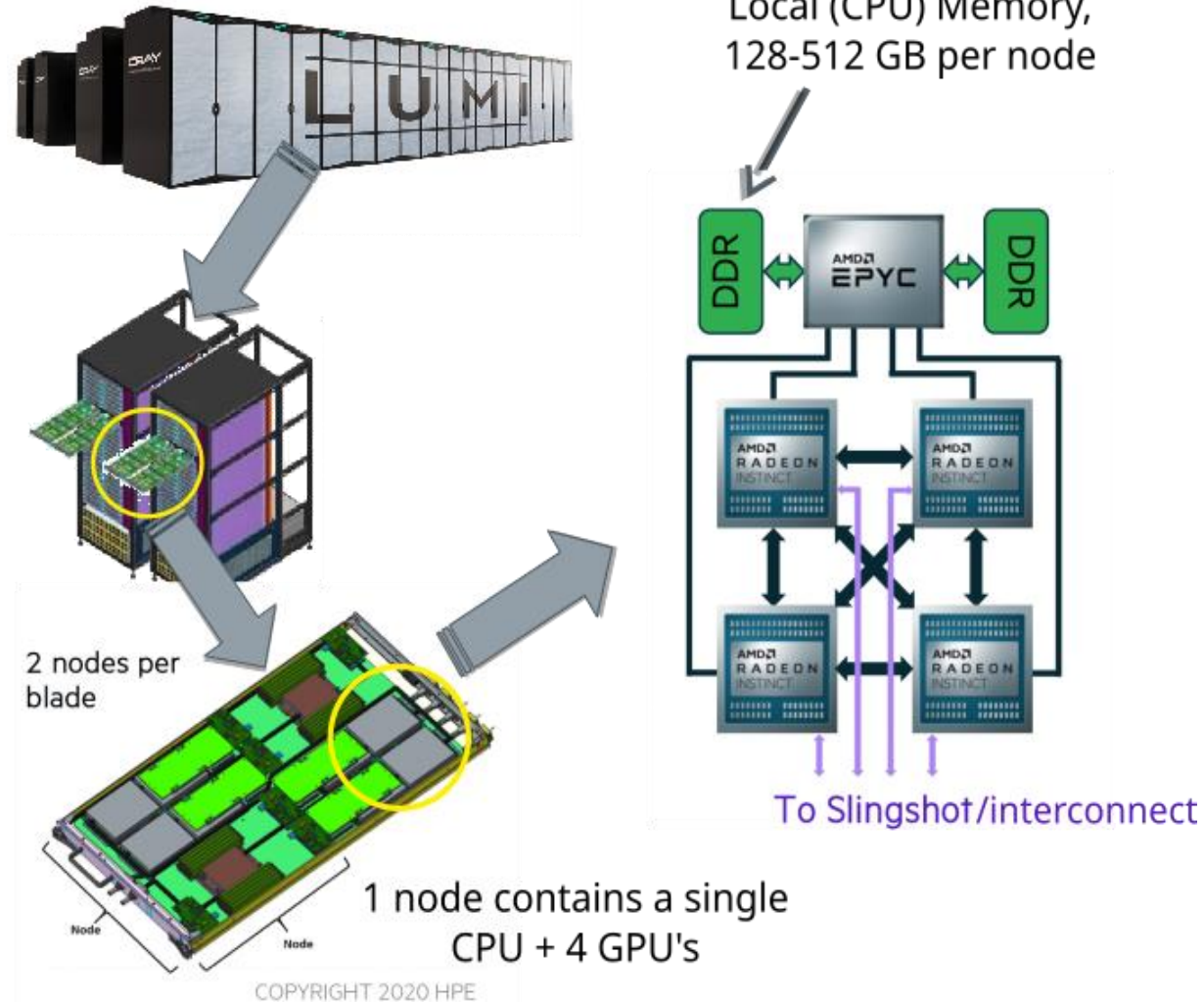
- (Possibly) Disk space



- (Possibly) GPUs



Case study: LUMI (GPU-partition)



Supercomputers

- Performance can be measured in floating point operations per second (flop/s)
- Supercomputers employ a "massively parallel" design
- Most of the calculation power usually comes from GPUs nowadays
 - GPUs provide better flops/watt
 - Other approaches also possible, based on different CPU and dataflow designs (e.g. Fugaku)

What are supercomputers used for?

What's being run on supercomputers?

- A large part nowadays dedicated to NLP/Large language models
 - Often utilize GPU's well, thus suitable for modern supercomputers
- DFT Simulations
 - GPAW
- Molecular dynamics
 - GROMACS, NAMD
- Weather modeling and forecasting
 - CESM
- Glacier modeling
 - Elmer

Examples: CESM

- "Community Earth System Model"
- One of the programs at IndySCC 2023
- Simulating global atmosphere, sea levels, land models etc.
- Simulating ~20 days of a given model took team Norppa ~20 hours of computing time

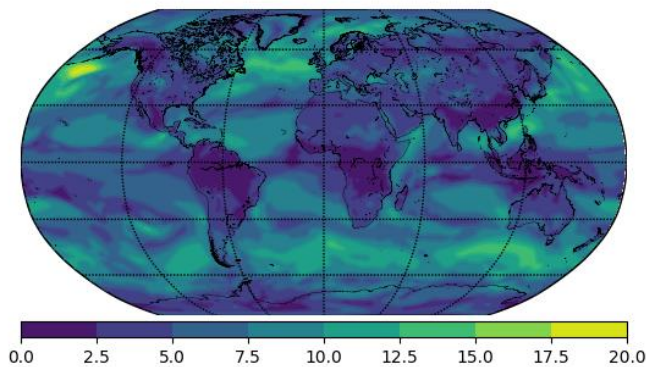


Fig. 1. Occurrence of wind-speeds over 10 m/s during a 5-day simulation

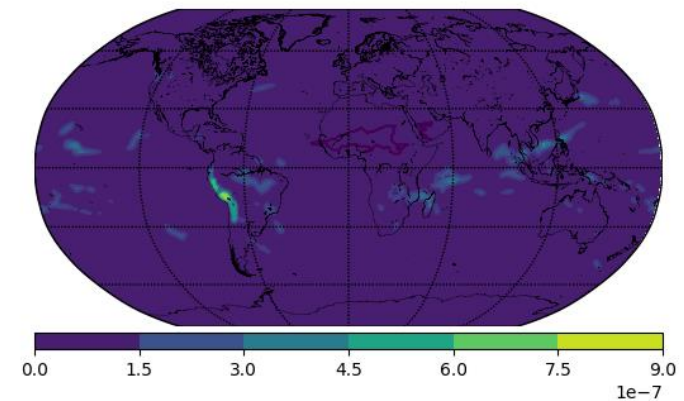


Fig. 2. Precipitation levels during a 5-day simulation

Examples: GROMACS



- Molecular dynamics package
- One of the applications at IndySCC '23
- Protein+Ligand solvent simulation as one of the exercises before the competition
 - Team Norppa got full points

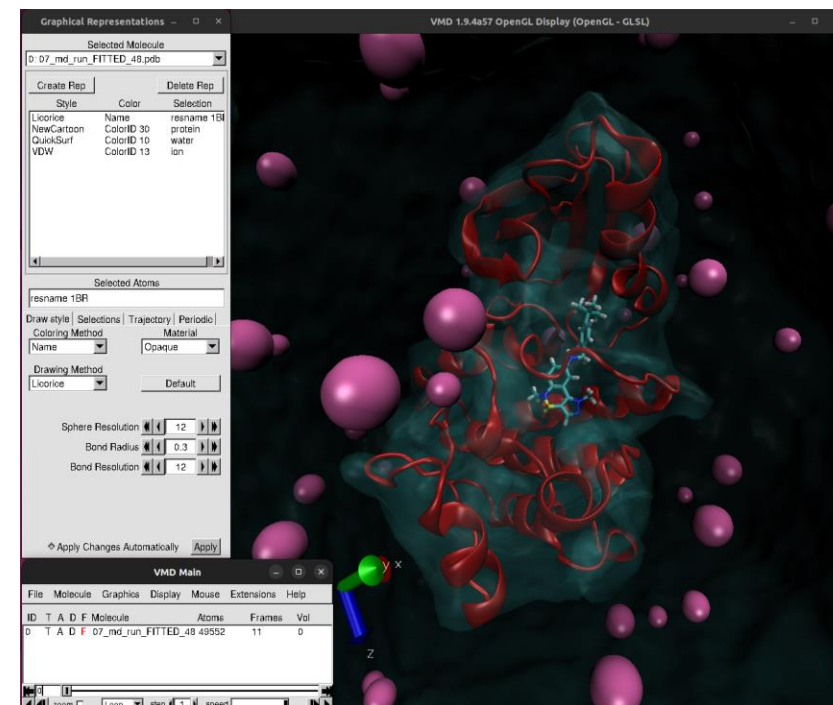
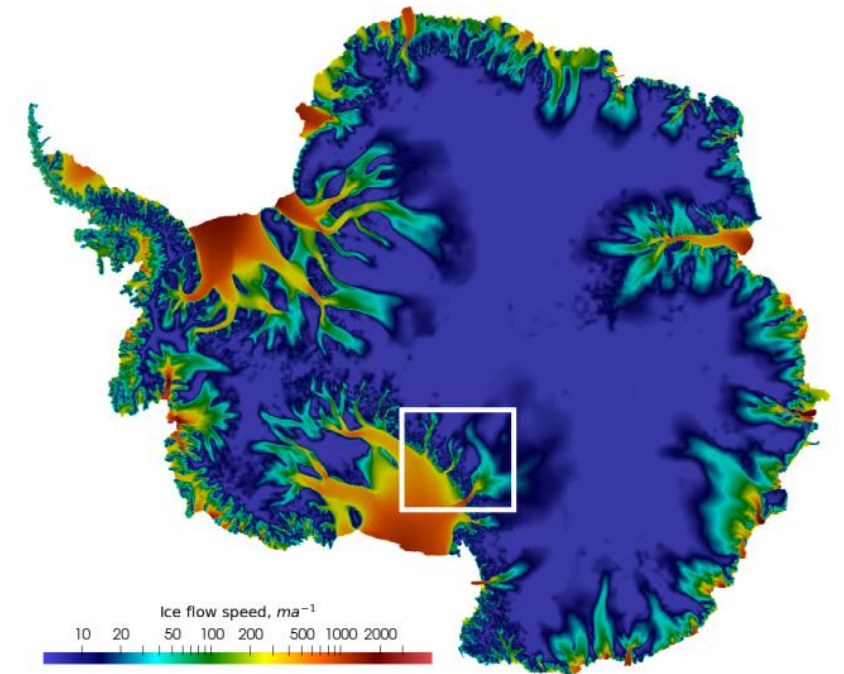


Fig. 2. A protein structure (red ribbon) with a ligand molecule (light-blue ribbon) surrounded by water molecules

Examples: Elmer(Ice)



- Multi-physics solver tool
- Finite-element method (FEM)
- Simulating e.g. Glaciers, electrical machines, permafrost
- Developed since 1995 at CSC



CSC's supercomputers and other services

Supercomputers in Finland

- Two national systems
 - Mahti and Puhti
- One system that is funded by the EuroHPC Joint undertaking
 - LUMI
- All are located in Kajaani
- Additional University clusters
 - Turso, Triton, etc.



Other CSC services

- Cloud systems
 - cPouta and ePouta
- Container orchestration services
 - Rahti
- Haka identity federation
- Funet
 - Finnish University and research network
 - Coordinates the eduroam-network in Finland

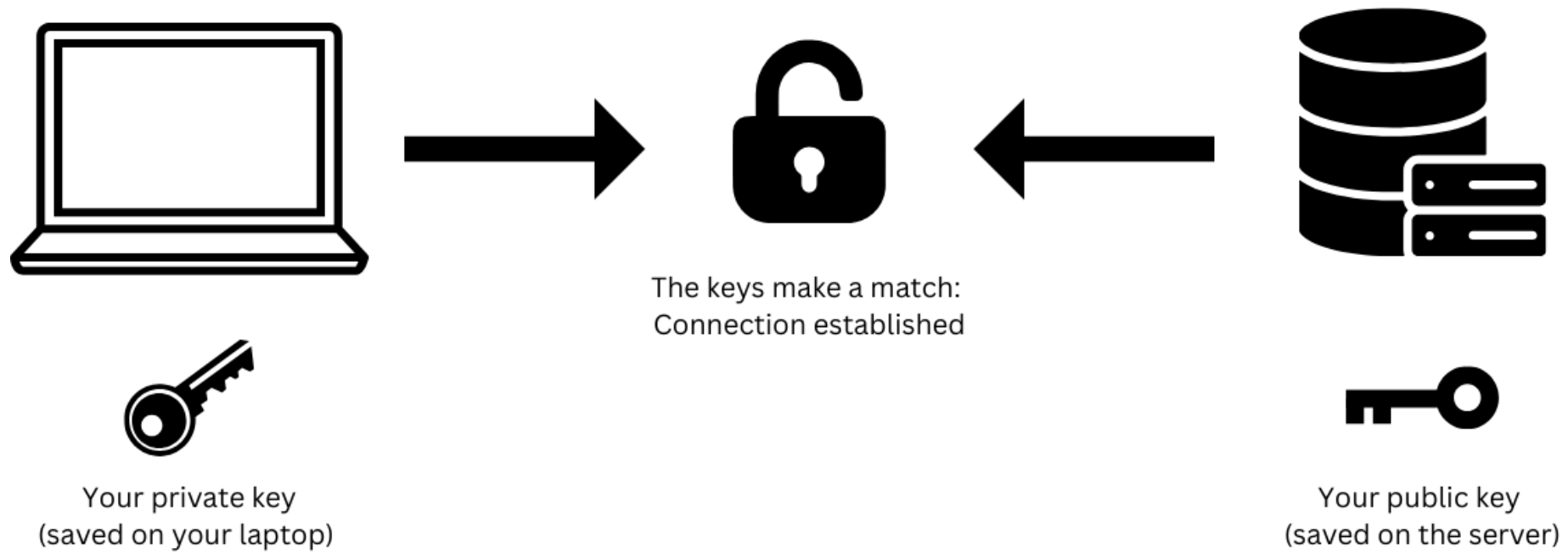
SSH Protocol and SSH Keys

The SSH Protocol

- Stands for "Secure shell"
- A protocol that enables secure connecting and file transfers over insecure networks
- Connections between computers only approved between trusted sources
- Message encryption and identification is based on SSH key-pairs

SSH Keys

- You can create local SSH key-pairs on your local computer to act as **identifiers**
- Each consist of a **Private key** and a **Public key**
 - **Important:** Never share the private key with anyone!
 - The public key can be shared (discretion is advised)
- If another computer has your public key as an authorized key, you can establish an SSH connection with it by using your private key



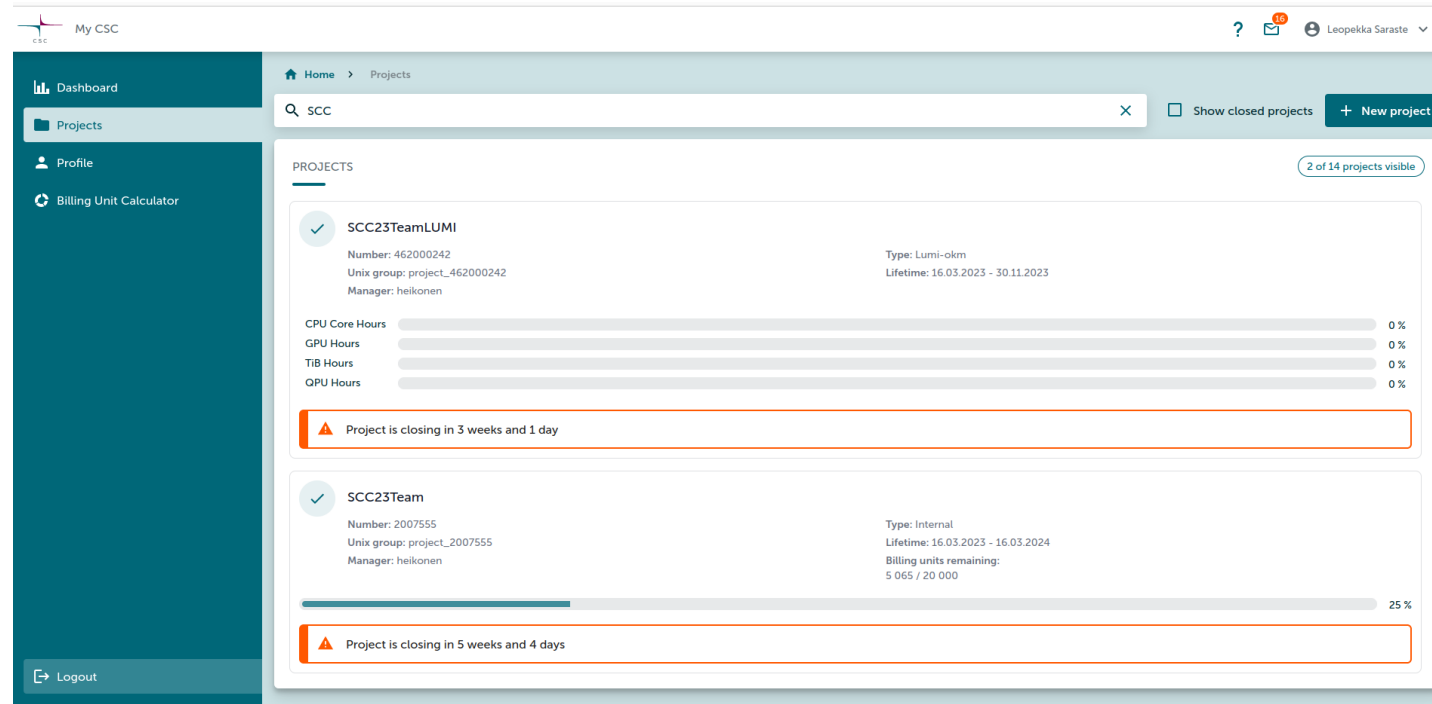
Create your own SSH Key pair for your laptop

- You can also use an existing one
- Instructions at:
<https://docs.csc.fi/computing/connecting/#generating-an-ssh-key-pair>
- Navigate on your computer to the `.ssh` folder in your home path and follow the instructions
- Give the key a fitting name and a passphrase

Logging into the Mahti supercomputer

Log into my.csc.fi

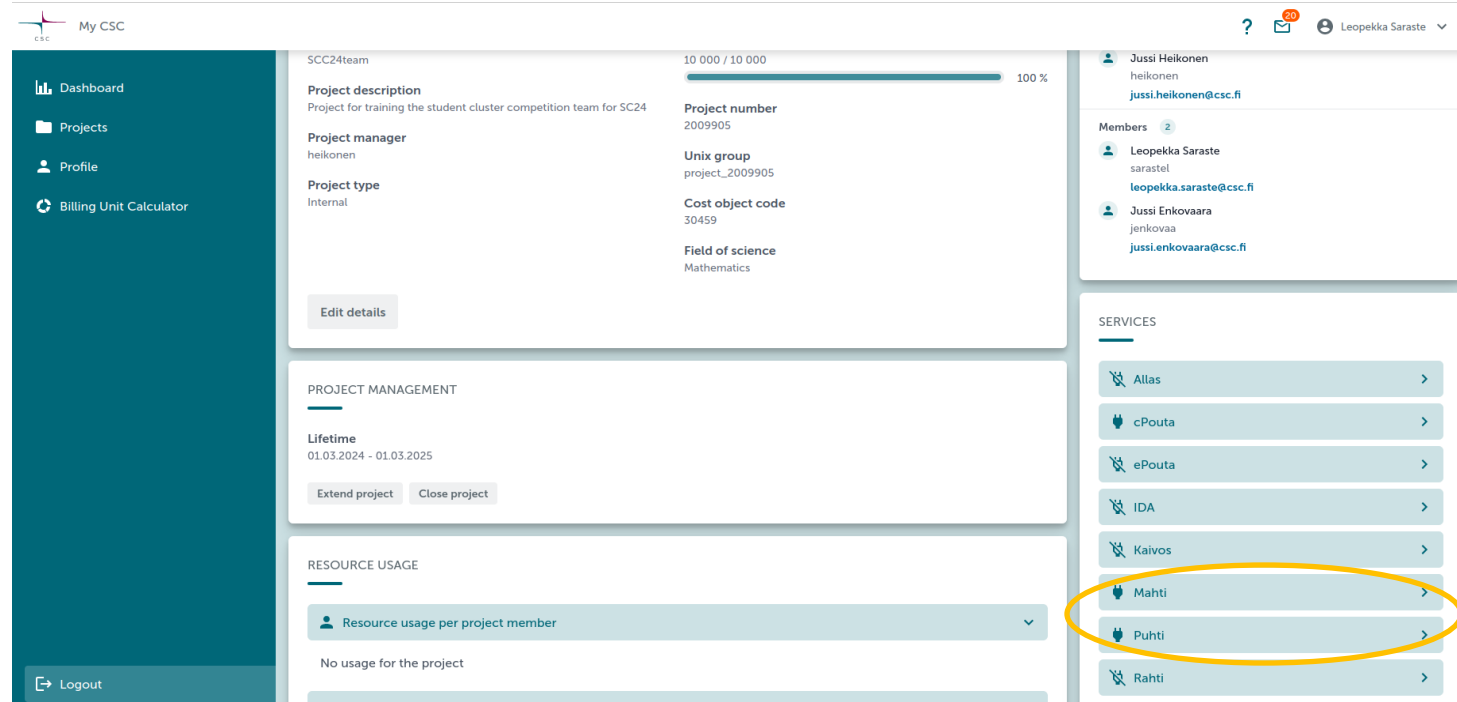
- Use e.g. Haka login or a CSC login
- Go to "Projects" in the left menu
- Take a look at the projects you have available



The screenshot shows the 'My CSC' web interface. On the left is a dark teal sidebar with a menu containing 'Dashboard', 'Projects' (selected), 'Profile', and 'Billing Unit Calculator'. At the bottom of the sidebar is a 'Logout' button. The main content area has a light blue header with a search bar containing 'SCC', a 'Show closed projects' checkbox, and a '+ New project' button. Below the header, the 'PROJECTS' section displays two project cards. The first card, 'SCC23TeamLUMI', includes details like Number (462000242), Unix group (project_462000242), Manager (heikonen), Type (Lumi-okm), and Lifetime (16.03.2023 - 30.11.2023). It features progress bars for CPU Core Hours, GPU Hours, TiB Hours, and QPU Hours, all at 0%. A warning banner indicates 'Project is closing in 3 weeks and 1 day'. The second card, 'SCC23Team', includes details like Number (2007555), Unix group (project_2007555), Manager (heikonen), Type (Internal), Lifetime (16.03.2023 - 16.03.2024), and Billing units remaining (5 065 / 20 000). It shows a progress bar at 25% and a warning banner indicating 'Project is closing in 5 weeks and 4 days'.

Gain access to Mahti

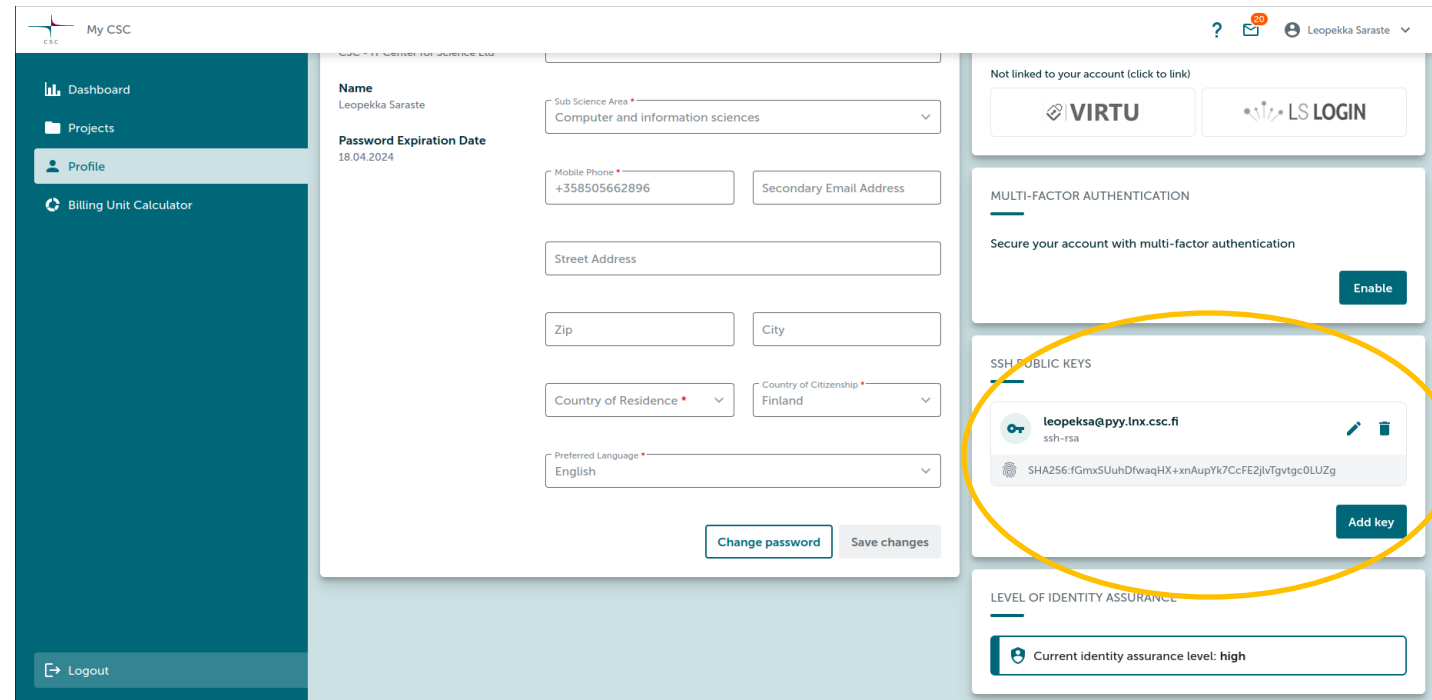
- Choose the project "SCC24Team"
- From the right hand, "service" menu, choose Mahti
- Click the checkbox to notify that you know the terms and conditions



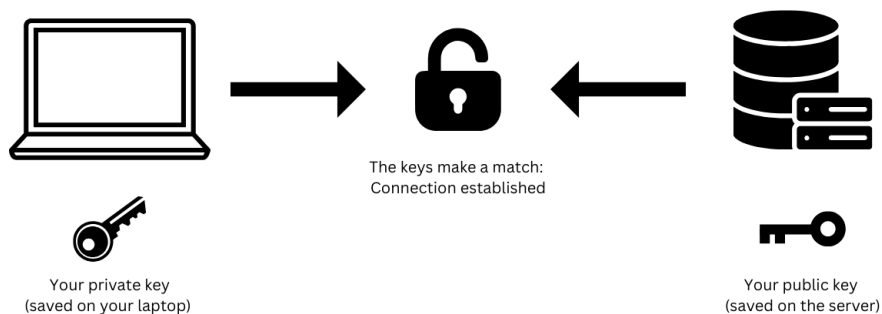
The screenshot displays the 'My CSC' web application interface. On the left is a dark teal sidebar with navigation links: Dashboard, Projects, Profile, and Billing Unit Calculator. The main content area shows details for the 'SCC24team' project, including a progress bar at 100%, project description, manager (heikonen), type (Internal), and various codes. Below this are sections for 'PROJECT MANAGEMENT' and 'RESOURCE USAGE'. On the right, a sidebar contains user information for Jussi Heikonen and a 'SERVICES' list. The 'SERVICES' list includes Allas, cPouta, ePouta, IDA, Kaivos, Mahti, Puhti, and Rahti. The 'Mahti' service is highlighted with a yellow oval. The user's name 'Leoekka Saraste' is visible in the top right corner.

Add your SSH key into my.csc.fi

- Next, go to "Profile"
- Add the ssh key you just created
 - Give it a fitting title



The screenshot shows the 'My CSC' user profile page. The left sidebar contains links to Dashboard, Projects, Profile, and Billing Unit Calculator. The main content area shows the 'Profile' section with fields for Name, Password Expiration Date, Sub Science Area, Mobile Phone, Secondary Email Address, Street Address, Zip, City, Country of Residence, Country of Citizenship, and Preferred Language. The 'SSH PUBLIC KEYS' section is highlighted with a yellow circle, showing an existing key for 'leopekka@pyy.lnx.csc.fi' with an 'Add key' button next to it. The 'LEVEL OF IDENTITY ASSURANCE' section shows the current identity assurance level as 'high'.



Log into Mahti!

- Open a new terminal
- Type in: `ssh <your csc username>@mahti.csc.fi`
- If all goes well, you're in!

Accessing our training GitHub through Mahti

1. Create an SSH keypair inside Mahti
2. Save the newly-created public key to your GitHub
 - In Github under "settings -> SSH and GPG keys"
3. Create a working folder for yourself in Mahti, under our SCC project
 - Type in "csc-workspaces" to find our SCC project path
 - Name it e.g. your username with the command `mkdir /scratch/project_2009905/$USER`
4. Copy the Student Cluster competition Github page into Mahti, with "git clone"



Thank you!



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linkedin.com/company/csc--it-center-for-science



github.com/CSCfi